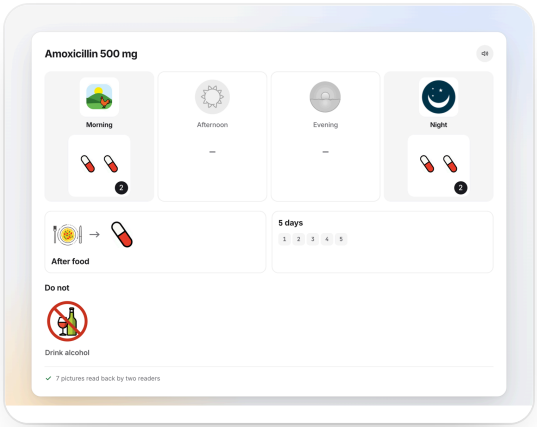
Visual discharge instructions, verified before the patient sees them.

A discharge sheet, typed or photographed, becomes picture cards a patient can follow. An image model draws every pictogram. Two other models have to read each one back correctly before it reaches a card.



Live concord-health.vercel.app, no sign-in
Code github.com/kinnuworks/concord
Video 1 minute 45 seconds, linked on the submission page

Where to look, by judging criterion

01
Technical execution

A five-step pipeline with our own fine-tune. Two models from different companies read the sheet, and any field they disagree on is flagged. Code lays out each card. FLUX.1 [dev] with a LoRA adapter we trained draws one fact per picture. Two vision models then describe the picture with no hint, a separate judge compares, and a misread picture is redrawn up to three times.

See `src/lib/gate.ts` `src/lib/readback.ts` `scripts/train.ts`

02
Innovation

A generated picture has to pass a reading test. Readers that were told nothing must say in their own words what it shows, with no list of answers to pick from. And the image model is never asked to count: it draws one pill, the check confirms it is exactly one, and code repeats it.

03
Use of image AI

Every pictogram on a card is generated, and the design follows from how an image model behaves. Our adapter, trained on 70 open pictograms, took pictures that read as flat pictograms from 21% to 78%, so a set of cards shares one look.

See the section "Where the image model works" on the site

04
Impact

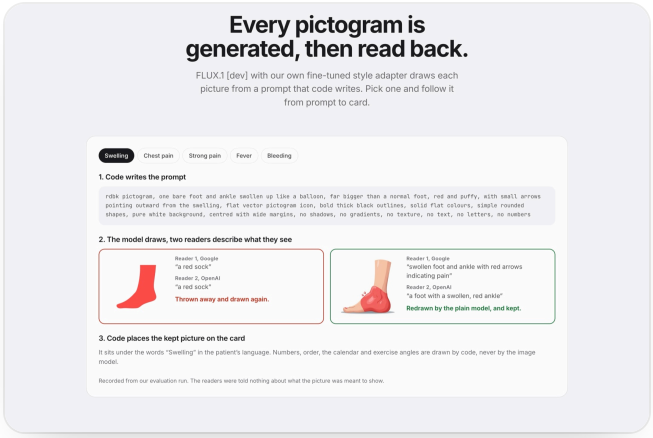
In one study, 78% of emergency patients went home with a gap in understanding their care (Engel et al., 2009). Concord gives a nurse or pharmacist a printable sheet the patient can talk through before leaving, in nine languages, and read aloud for people who would rather listen.

05
Design and demo

The site opens on finished cards. A photographed sheet turns into cards in a few seconds, the language switches in one click, and the cards print on two clean pages. The video shows all of this on the live site.

Try it in two minutes

1. Open **concord-health.vercel.app**. Finished cards are already on screen.
2. Click **A photo of a printed sheet**, then **Make the cards**. The sheet is read live by two models.
3. Change the language to Spanish, press the speaker on a card, then **Print the cards**.
4. Scroll to **Where the image model works** and click through the five examples. Each shows the prompt, a picture the readers misread, and the one that was kept.



A recorded run. Asked for a swollen foot, the model drew what both readers called "a red sock". It was thrown away and drawn again.

What we measured

83 of 168 An image model on its own whole-dose pictures showed the wrong number of pills when the model was asked to draw the dose in one picture.

58 of 60 With Concord single-pill pictures passed the read-back on the first try. Code repeats the checked pill, so the count on a card is exact.

21% to 78% With our adapter pictures both readers called a flat outlined pictogram, 270 pictures each way.

100 of 100 With Concord written test instructions read correctly by the two sheet readers.

540 pictures were drawn for a fixed set of 100 instructions and every one was read by both readers. The scripts are in `scripts/` and the raw readings in `training/eval/`.

Built during the hackathon

- A LoRA style adapter for FLUX.1 [dev], trained on 70 OpenMoji pictograms in 1,000 steps.
- The read-back check: two readers, a judge, redraws, and a code-drawn pictogram when nothing passes.
- A 100-instruction test set and the 540-picture evaluation behind every number above.
- Card labels in nine languages: English, Spanish, Chinese, Tagalog, Vietnamese, Arabic, Hindi, Telugu and Tamil.
- The live site, with photo input, read-aloud and print. Every commit falls inside the hackathon window.